

Cheap Signals, Costly Proof: Award-Layer Triage in Cartel Enforcement

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Abstract

Cartel cases begin under thin information. Agencies often see award records long before they recover the bid-level evidence needed for proof. We ask whether that cheap layer can guide where costly forensics should start. Using São Paulo’s BEC procurement platform (2009–2019), we build a *frequent-loser* screen from award records alone: among zero-win firms, participation intensity ranks loser-side cartel-adjacency risk. Computed from 2009–2016 participation, the screen discriminates later-adjudicated CADE cobidders in 2017–2019. On the same target, it achieves comparable discrimination to a seven-feature Imhof–Wallimann bid-distribution pipeline and adds non-redundant signal in combination. Used as a gatekeeper, the award-layer screen cuts the bid-microdata pool for the forensic stage by 83% while still flagging 131 of 193 adjudicated cobidders. The statistic creates suspicion, not proof: it ranks cartel-adjacent firms and environments for forensic attention without adjudicating membership or estimating damages.

Keywords: cartel screening, public procurement, award-layer data, enforcement triage, incomplete observability, bid rigging

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1. Introduction

Cartel enforcement begins before cartel proof exists. Agencies must decide where to spend investigative resources while the evidence needed for liability is still costly to recover. The economics of law enforcement (Becker, 1968; Stigler, 1970; Baker, 2003; Harrington, 2008) prescribes that when observation is costly, an initial screen should use information already in the agency’s hands and reserve costly evidentiary stages for the candidates the screen ranks highest. Procurement-cartel detection usually begins one layer later. The empirical industrial-organization literature reads coordination off the distribution of submitted bids (Porter and Zona, 1993; Bajari and Ye, 2003; Imhof, 2019; Wallimann et al., 2023; Chassang and Ortner, 2022): per-bidder amounts, within-tender dispersion, bid rankings, and sometimes bid histories. Many enforcement environments routinely expose only a thinner layer: who participated, who won, what was bought, and at what price. The bid-by-bid record may exist, but it sits in operational files that have to be recovered through a case-specific forensic request.

The question here is what antitrust enforcement can do at that earlier layer. The answer is not cartel proof. It is evidentiary triage: a cheap award-layer ranking of firms and procurement environments that indicates where bid-layer forensics should begin. We study this architecture on Sao Paulo’s BEC procurement platform (2009–2019, 1,654,401 tender-items).

The screen uses four award-record fields: participant identity, winner identity, item code, and negotiated price. It flags *frequent losers*: firms with zero wins whose participation intensity among always-loser firms lies above a median plus 1.5 interquartile-range rule. The logic is simple. In a cover-bidding arrangement, loser-side firms are deployed to create the appearance of competition while avoiding the win. The wins-zero filter isolates the side of the bidding pool the award layer can plausibly see; participation intensity within that stratum ranks exposure to the cartel’s deployment problem. The binary frequent-loser rule is a coarsening of that rank, not a claim of legal membership.

That scope determines the validation target. We do not ask whether award records identify cartel members for sanction. They do not contain the evidence for that claim.

We validate instead against loser-side cartel adjacency: firms in the always-loser stratum that bid alongside direct defendants in adjudicated CADE procurement-cartel cases. Direct defendants are legal-membership anchors and are often winner-side; cobidders are the loser-side footprint the award-layer screen is designed to rank. Procurement-cartel scholarship documents the same asymmetry: when collusion takes the form of bid rotation or market allocation, winners are designated by allocation rules and rotate across tenders, while cover bidders are the assigned losers (Marshall and Marx, 2012; Kawai and Nakabayashi, 2022). Stronger discrimination against cobidders than against direct defendants is therefore a design signature, not a fallback target.

Five empirical implications organize the exercise. The screen should discriminate adjudicated cobidders inside the always-loser population, including when the score is formed before the labels are observed. As a loser-side construct, it should perform worse against direct CADE defendants than against cobidders. If cobidders are more than active losing firms, their firm- and bid-level profiles should resemble credible cover bidding: deployment near direct defendants, concentrated market portfolios, plausibly close losing bids, and dispersion consistent with rotation across cover roles. If award and bid records carry different information, the award-layer statistic should complement rather than replace bid-distribution methods. Finally, if the architecture has enforcement value, sequencing award-layer screening before bid-layer forensics should reduce the bid-microdata pool while preserving most recall against adjudicated cobidders.

The results line up with those implications. The main institutional payoff is the sequential gatekeeper exercise: placing the award-layer screen ahead of bid-layer forensics cuts the bid-microdata pool the forensic stage must process by 83% while still recovering 131 of 193 adjudicated cobidders. Discrimination is out-of-sample, not retrospective fitting: built on 2009–2016 participation and evaluated on 2017–2019 labels, the screen ranks later-adjudicated CADE cobidders at firm-level AUC 0.864. On the same cartel-adjacency target, it reaches discrimination comparable to a seven-feature Imhof–Wallimann bid-distribution pipeline and adds non-redundant signal when the two are combined (+0.035 AUC, DeLong $p = 0.014$). A four-field award-layer statistic does not dominate bid-distribution forensics. The two layers are complements: the cheap layer tells the agency

where the costly layer should be applied first.

The cobidder evidence narrows the interpretation. Within the frequent-loser stratum, cobidders face more unique winners, bid more often alongside direct CADE defendants, and concentrate activity in narrower item-group portfolios than other frequent losers. The LANCES export lets us look one layer deeper: cobidders bid closer to winners and show higher within-firm dispersion in the bid gap, with both signs surviving a multivariate logit that controls for participation intensity. These facts do not prove cartel membership. They address the most direct objection to the validation exercise: cobidders are not merely high-volume losing firms; their loser-side profile is consistent with credible cover bidding in the modern cover-bidding literature ([Asker, 2010](#); [Marshall and Marx, 2012](#)).

The price evidence is subordinate. Frequent-loser presence is associated with a $+3.6\% - +7.7\%$ higher conditional log negotiated unit price across four broad-sample estimators, with the positive imprint concentrated in the highest tender-value segment. Under overlap restriction the aggregate sign reverses, but the highest segment remains positive. We report this as corroboration that the screen tracks environments where collusion-relevant behavior matters, not as a damages estimate or a causal effect on prices. The main claim does not rest on either sign of the price coefficient.

The contribution is to change where familiar tools enter the enforcement sequence. Enforcement-design models explain why costly investigation should be sequenced ([Becker, 1968](#); [Stigler, 1970](#); [Baker, 2003](#); [Harrington, 2008](#)); bid-distribution screens show how to detect suspicious bidding once bid microdata are available ([Porter and Zona, 1993](#); [Bajari and Ye, 2003](#); [Imhof, 2019](#); [Wallimann et al., 2023](#); [Chassang and Ortner, 2022](#)); bid-coordination tests, including [Conley and Decarolis \(2016\)](#)-style coalition inference, evaluate collusion conditional on suspected partitions or rich bidding behavior inside an already suspected environment; and cover-bidding studies characterize loser-side roles after adjudication, leniency, or discovery ([Porter and Zona, 1999](#); [Pesendorfer, 2000](#); [Asker, 2010](#)). A more recent procurement-enforcement-design literature studies how rules, discretion, and bureaucratic selection at the award stage shape outcomes ([Decarolis, 2014](#); [Decarolis et al., 2025, 2020](#)); this paper studies what an enforcement screen can do with the information those settings take as their starting point. We study the earlier stage

at which none of those objects is yet available to the agency at low cost: the empirical question is whether award records alone can rank loser-side cartel-adjacent firms well enough to decide where bid-layer forensics should begin.

Three boundaries discipline the claim. The screen creates suspicion, not legal proof. The validation target is cartel-adjacent cobidding, not cartel membership. The price imprint is descriptive, not a damages calculation. These limits define the stage of enforcement studied here rather than apologize for it. The resulting claim is deliberately narrower than cartel detection and broader than a new statistic: when proof is costly and suspicion can be generated from records the agency already holds, the policy question is not whether the cheap layer can decide liability—it cannot—but whether it can discipline the allocation of costly forensic attention. In this setting it can. The award layer supplies a legally limited but empirically informative triage stage; the bid layer remains where proof is assembled.

2. Costly Proof and Observable Awards

2.1. Screening, Investigation, and Proof

Antitrust agencies rarely begin with the evidence they need to prove a procurement cartel. They begin with coarse records, complaints, leniency leads, procurement anomalies, or repeated interactions that make some firms and markets worth a closer look. Legal proof comes later, after the agency recovers bid histories, communications, leniency testimony, and other case-specific evidence sufficient to support adjudication. The first institutional problem is therefore not classification once rich evidence is available; it is allocation of costly proof-producing effort.

That distinction defines the role of an award-layer screen. The relevant performance criterion is not whether the screen establishes liability, but whether it improves the allocation of costly forensic attention. Bid-distribution methods ask whether the submitted bids look suspicious once per-bidder bid amounts are available. An award-layer screen asks an earlier question: before recovering those bid amounts, can the agency rank the firms and procurement environments where bid-layer forensics is most likely to pay off?

Bid-distribution forensics remains indispensable at the evidentiary stage; award-layer triage helps decide where that stage should start.

The institutional sequencing problem has a substantial law-and-economics tradition. The Becker–Stigler framework treats enforcement intensity as a choice variable under costly observation (Becker, 1968; Stigler, 1970). Subsequent work formalizes screening as a first-stage filter that allocates investigative effort over a population of potentially infringing firms (Harrington, 2008; Baker, 2003): when proof is expensive and conduct is partially observable, an inexpensive ranking step on routinely available data is the rational starting point of investigation. The award-layer construction in this paper instantiates that filter. It scores firms using information the agency already has, before deciding whether to mobilize the costly bid-layer evidentiary stage.

What counts as proof in a procurement-cartel case is more substantial. The agency must establish a coordinated bid-rigging arrangement that produced anticompetitive harm, typically through evidence of communication, allocation rules, or bid patterns inconsistent with independent decision-making. Brazilian competition law treats such conduct as an infraction under Lei 12.529/2011; comparable frameworks in the United States and the European Union also place the agreement element and its effect at the center of the case (Hovenkamp, 2005). The screening object analyzed here does none of this. It produces a ranking that helps the agency decide where to seek the evidence those legal frameworks require.

2.2. The BEC Enforcement Environment

São Paulo’s BEC platform is the right environment for this design because it places the routine record layer on which triage can run, the forensic-recoverable bid records, and external CADE adjudication labels in the same procurement system. It covers 1,654,401 tender-items during 2009–2019 and preserves a contract-award envelope for each tender-item. Procurement-cartel cases are adjudicated by CADE, the Brazilian competition authority, supplying external legal anchors for validation. The combination lets us study the paper’s central allocation problem: whether information available before bid recovery can rank where costly forensic effort should begin.

Two procurement modalities dominate. *Convite* is a sealed-bid invitation under Lei 8.666/93 with a statutory three-bidder quorum and a low value cap; *Pregão* is a reverse electronic auction under Lei 10.520/2002 with progressively lower real-time offers and no minimum-bidder rule. The modalities differ in bidding rule and observability regime: *convite* is closer to an award-only environment, while *pregão* generates a richer real-time interaction footprint. We treat this contrast as scope information, not causal identification. The common feature matters more for the architecture: both modalities leave an award layer that can be queried before bid-layer evidence is recovered.

Cartel enforcement in Brazilian procurement is divided across institutions. CADE, the federal antitrust authority, investigates and adjudicates anticompetitive conduct including bid rigging. Federal and state audit courts—TCU at the federal level, TCE-SP at the state level—supervise procurement regularity and refer cases to CADE or to the Public Prosecutor’s Office (*Ministério Público*) when collusion is suspected. The screen analyzed here lives in the institutional space of CADE’s bid-rigging caseload: it ranks firms whose award-layer behavior warrants the costly bid-layer recovery a CADE investigation would require. The same record layer is, however, already routinely consumed by audit-court and prosecutorial staff during procurement oversight (OECD, 2019), which is one reason the architecture is institutionally implementable rather than purely academic.

2.3. Award Layer and Forensic-Recoverable Bid Layer

BEC exposes two analytically distinct layers. The *award layer* carries the fields an enforcement analyst can query routinely: participant identity, winner identity, item code, negotiated price, procuring unit, year, and procurement modality. The *forensic-recoverable bid layer* carries per-bidder bid amounts, bid timestamps, and bid-revision sequences. Those bid records exist in the BEC operational system, but they are not exported to the ordinary analytical warehouse. They must be recovered through an administrative LANCES export: feasible for evaluation and case work, but costly relative to querying the award layer. OECD reviews of Brazilian procurement infrastructure describe this kind of administrative latency as a practical constraint on routine monitoring (OECD, 2019). The constraint is not data non-existence; it is the cost and timing of

turning operational bid files into analyzable forensic evidence.

Figure 1 is not an accuracy horse race; it is an information-cost diagram. The left side reports the bid-layer features required by the Imhof–Wallimann-style forensic benchmark: within-tender coefficient of variation, dispersion across tenders, skewness, kurtosis, normalized spread, log min–max, and second-lowest bid distance. The right side reports the four fields used by the award-layer screen. The empirical question is whether the right-hand layer can rank where the left-hand layer should be opened first; the bid-layer features remain the forensic-stage benchmark, recovered through LANCES only after triage has selected where to look.

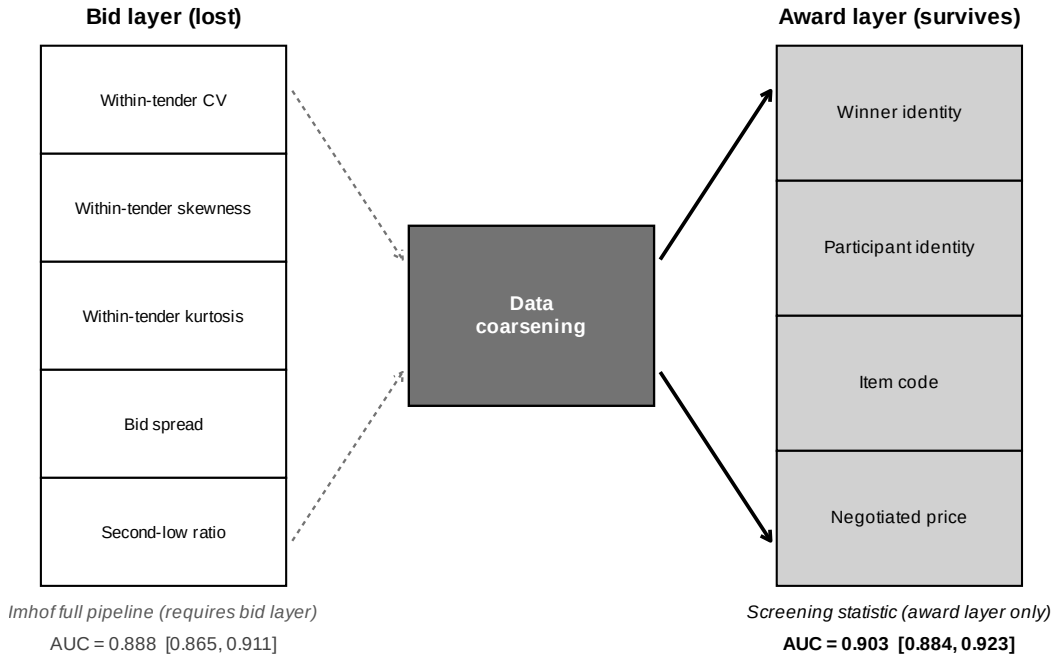


Figure 1: Information layers in procurement-cartel enforcement. Bid-distribution forensics requires per-bidder bid amounts and within-tender bid moments; the award-layer screen uses participant identity, winner identity, item code, and negotiated price. The empirical question is whether the right-hand layer can rank where the left-hand layer should be recovered first.

The same institutional distinction appears outside BEC. In the United States, FPDS-NG preserves federal contract awards as routine administrative records, while bid files sit with contracting agencies and are typically recovered through case-specific process. In the European Union, TED mandates publication of contract-award notices, but per-bidder bid amounts are preserved heterogeneously across member-state systems; a recent

procurement-screening review reports that fewer than half of surveyed systems make bid amounts retrievable at the analytical-warehouse layer (Sánchez Graells, 2019). These comparisons are not external validation of the BEC estimates. They establish a narrower point: the institutional distinction between routine award records and costly bid files is recognizable outside this setting.

The bid-distribution literature presumes the left-hand side of Figure 1. The variance and dispersion screens of Abrantes-Metz et al. (2006) use within-tender bid moments; the machine-learning extensions of Imhof et al. (2018); Imhof (2019); Huber and Imhof (2019); Wallimann et al. (2023) apply richer functions to the same per-bidder bid amounts. The coalition-inference apparatus of Conley and Decarolis (2016) requires bid-level histories. These methods are not portable to the right-hand side because their inputs are not yet recovered there, not because the underlying logic fails. The paper’s question is therefore not whether to use bid-distribution moments—we use them at the forensic stage—but whether award-layer information can decide where to recover those moments first.

2.4. CADE Labels and the Cartel-Adjacency Target

CADE adjudications provide the legal anchors for the validation exercise. CADE enforces Brazilian competition law under Lei 12.529/2011, and its procurement-cartel decisions identify firms legally named in cartel cases. The portfolio linked to BEC contains 12 adjudicated procurement-cartel cases with 65 firm-defendants, of which 47 are active in BEC during the sample window. Final decisions span 2015–2025, with process-to-ruling intervals of 4–16 years. That lag is central to the screening interpretation: an award-layer screen has value only if it can move suspicion toward forensic investigation before the completed adjudication record exists.

The adjudication process behind those labels has a standard structure. An investigation is opened by CADE’s *Superintendência-Geral* (SG), often following a leniency application, a *Termo de Compromisso de Cessação* settlement, or a referral from another enforcement body. The SG issues a recommendation; CADE’s Tribunal reviews and rules. Cartel cases are heavily leniency-driven: the leniency provisions of Lei 12.529/2011, operationalized through CADE regulations, supply most of the direct-defendant identi-

fications in our portfolio. This matters for what direct defendants are. They are firms whose membership was established through legal process, often via the cooperation of another cartel. The award-layer screen has no equivalent legal authority and produces no analogous identification.

The adjudication record creates two distinct empirical objects. Direct CADE defendants are legal membership anchors: they locate cartel-affected environments but are not the natural target of a loser-side award-layer statistic, because in procurement cartels they often occupy winner-side or allocation roles. Cobidders are different. They are always-loser firms that appear alongside direct defendants in BEC tender-items; they are indirect labels, and we do not treat them as proof of cartel membership. They are the loser-side adjacency object the award layer can plausibly recover: firms that lose, continue to appear, and are observed near legally adjudicated cartel anchors.

This distinction is not a fallback after failed defendant classification. It follows from the information layer. Award records contain participation and outcome information; they do not contain agreements, communications, or the tender-specific conduct needed for liability. A statistic built from zero wins and repeated participation should therefore be evaluated against loser-side exposure, not legal membership. Weak performance against direct defendants is a scope check rather than a failure: it shows that the statistic is not a generic cartel-membership detector.

The rest of the analysis follows that boundary. We evaluate whether award records rank cartel-adjacent loser-side firms well enough to allocate bid-layer forensic effort. The screen remains a triage device; it does not adjudicate liability.

3. A Loser-Side Screen for Cartel Adjacency

3.1. The Participation Primitive

The award layer cannot reveal why a firm lost. A losing bid may be sincere, poorly priced, capacity constrained, exploratory, strategic, or coordinated; the analyst querying award records cannot distinguish these from one tender’s outcome. What the award layer can observe is thinner: which firms appeared, which firms won, and how often the same

firm kept appearing without ever winning. The primitive used here is therefore not a behavioral type. It is an observable loser-side exposure measure built from participation and zero wins. In the BEC data the relevant conditioning set contains 16,843 always-loser firms.

Zero wins alone is not enough. Many firms never win for ordinary competitive reasons. They may be too small or poorly matched to a particular item; they may face binding capacity constraints in the periods when they would otherwise have won; they may be using procurement participation for market intelligence, eligibility maintenance, subcontracting positioning, or relationship-building with procuring units. Without an additional discriminating signal, the always-loser stratum is too heterogeneous for triage. The screening signal is persistence inside that stratum: among firms that never win, how often do they keep showing up?

The maintained condition is monotonic rather than classificatory: within the zero-win stratum, cartel-deployed losing participants, if present, should participate more persistently than ordinary losing firms, because their function in the cartel—supplying the appearance of competition while leaving the designated winner in place—pays off through repeated deployment in cartel-targeted tenders. Under that condition, participation intensity inside the zero-win stratum, implemented as $\log(1 + \text{tenders_count})$, is a monotone ranking statistic for loser-side cartel-adjacency risk. The appendix formalizes this condition in a sparse screening framework; the main text uses it only to discipline the ranking interpretation.

This primitive does not identify cover bidders or cartel members. It does not establish agreement, liability, or damages. It defines a first-stage ranking: among firms the award layer only observes as losers, which ones appear persistently enough to justify bid-layer forensic attention? The binary rule in subsection 3.2 is only an operational coarsening of this continuous ranking.

3.2. The Binary Rule

The continuous ranking is the economic object. Within the always-loser stratum, we score firms by $\log(1 + \text{tenders_count})$, which orders repeated zero-win participation

under the monotonicity condition stated in subsection 3.1.

A binary frequent-loser flag is useful for a different reason. Enforcement programs need more than a rank: agencies must decide which firms enter a survivor pool, which records justify a costly bid-microdata recovery, and where a forensic queue begins on Monday morning. The cutoff converts the continuous score into an administrable first-stage list, with the auditable property that two analysts applying the same rule to the same participation distribution will obtain the same survivor pool.

For the discrete rule, we use the median plus $1.5\times$ interquartile range of the always-loser participation distribution. In BEC this gives a cutoff of 14 tender-items with zero wins and flags 2,735 frequent losers. The rule is fixed from the participation distribution before any CADE label enters the analysis and requires no bid amounts, bid ranks, or within-tender dispersion measures. The cutoff therefore inherits the low-information property that motivates the architecture.

The number 14 has no independent legal or structural status. It does not divide cartel firms from non-cartel firms, and it does not make participants below the threshold competitive or those above it liable. It is an auditable implementation of a continuous ranking, calibrated to the spread of zero-win participation in the data rather than to any structural parameter of the bidding environment.

Agencies could implement the same architecture differently. A capacity-constrained top- k list would draw a fixed number of firms from the highest-ranked end of the continuous score; a budget-calibrated threshold would set the cutoff to match the bid-microdata recovery capacity of an investigative team; a dynamically refreshed rule would update the threshold as new participation records arrive. The paper reports both the binary flag and the continuous score where the analysis allows, so that the validation in Section 4 can be read as evidence on the ranking architecture rather than on any one cutoff. Threshold manipulability by firms aware of the rule is a practical concern about the binary implementation, not a conceptual collapse of the architecture, and we return to it under adversarial adaptation.

3.3. Populations and Legal-Economic Role

The empirical design turns on keeping five populations separate. The broad BEC universe defines where procurement participation is observed. The always-loser stratum defines the side of the bidding pool that the award layer can rank. Frequent losers are the administrable subset flagged by the screen. Direct CADE defendants are legal membership anchors from adjudicated cases. Cobidders are always-loser firms observed alongside those anchors. These populations overlap in the data, but they do not play the same legal or empirical role.

Table 1 fixes those roles. The screen operates only inside the always-loser stratum. Frequent-loser status is an operational flag, not a legal category. Direct defendants locate cartel-affected environments, but a zero-win loser-side statistic is not designed to classify them. Cobidders define the cartel-adjacency target for validation: they are firms that lose, keep appearing, and are observed near adjudicated cartel defendants. That label is informative for validation but is not proof of agreement, membership, or liability.

Table 1: Populations and legal-economic role. Frequent losers are an operational screen; direct defendants are legal anchors; cobidders are cartel-adjacent loser-side labels. The table separates legal membership from screening exposure.

Population	Size	Role in the analysis
All BEC participants	41,444	Firms participating in BEC tenders during 2009–2019; the outer procurement universe.
Always-losers	16,843	Firms with zero wins; the loser-side stratum within which the award-layer screen ranks firms.
Frequent losers	2,735	Always-losers above the participation cutoff; the discrete first-stage screen.
CADE direct defendants in BEC	47	Firms named in CADE procurement-cartel rulings and active in BEC; the legal membership anchors.
Cobidders in the always-loser stratum	193	Always-losers that participated alongside direct defendants; the cartel-adjacency target for validation.

This separation is what makes the validation design legally limited rather than opportunistic. The paper does not collapse defendants and cobidders into one cartel class. Frequent-loser status is not cartel membership; cobidder status is not legal membership; direct-defendant status is the legal-anchor role, not the screen target. The screen ranks

loser-side exposure; CADE labels anchor the validation step.

Section 4 therefore validates the ranking, not the legal status of the ranked firms. It asks whether the frequent-loser ranking concentrates on the cartel-adjacent loser-side population at rates above what participation volume alone would imply.

4. Validating the Loser-Side Ranking

This section validates the loser-side ranking, not the legal status of ranked firms. The target is loser-side cartel adjacency: always-loser firms observed alongside direct CADE defendants in adjudicated procurement-cartel environments. That target follows from the data layer. Award records observe participation and outcomes; they do not observe agreements, communications, or tender-specific conduct. The question is therefore not whether the screen proves cartel membership, but whether it concentrates the loser-side firms most plausibly worth bid-layer forensic attention.

We organize the evidence as a validation architecture, with one exercise per threat to interpretation. The conservative adjudication benchmark (subsection 4.1) anchors the target in cases resolved before the end of 2020, reducing reliance on later case outcomes. The temporal holdout (subsection 4.2) asks whether the ranking discriminates later-adjudicated cobidders when the score is fixed before their labels exist. The participation-volume and leakage audit (subsection 4.3) asks whether frequent losers are overrepresented near defendants beyond what participation volume mechanically predicts, and whether the result survives stratification on participation history. The direct-defendant asymmetry (subsection 4.4) asks whether the statistic is mistakenly behaving like a generic cartel-membership detector; for a loser-side screen, it should not.

The section therefore evaluates the ranking against the threats that would make it uninformative for enforcement: mechanical exposure through participation volume, label leakage, retrospective fitting, and legal overclaiming. The performance criterion is not proof. It is whether award records rank cartel-adjacent loser-side exposure well enough to allocate costly bid-layer forensic effort.

4.1. A Conservative Adjudication Benchmark

The first benchmark is deliberately restrictive. We use only CADE procurement-cartel cases adjudicated before the end of 2020, leaving 4 cases, 30 firm-defendants active in BEC, and 210 always-loser firms that bid alongside at least one of those defendants. Of those cobidders, 108 are frequent losers. Because both the cartel finding and the defendant list precede the construction of the screen, this benchmark anchors the initial validation target in adjudications available early relative to the temporal-holdout exercises that follow. The restriction trades sample size for legal-timing discipline.

The main threat is mechanical exposure. Firms that participate often have more opportunities to appear near any defendant, regardless of cartel adjacency. Raw co-bidding rates would therefore overstate what the screen contributes. The relevant question is whether frequent losers are overrepresented near adjudicated defendants beyond what their participation volume alone would mechanically generate.

We address that concern with a participation-stratified permutation. Within quartiles of participation volume, we randomly permute cobidder status 1,000 times and recompute the frequent-loser share among placebo cobidders. The procedure preserves the exposure created by participation intensity while breaking the specific adjacency to CADE defendants. The observed co-bidding rate is 3.95%, against a permutation mean of 1.24%, an excess ratio of 3.2 $\times$ with $p < 0.001$. The ROC AUC against the same conservative ground truth is 0.748 (95% DeLong CI [0.713, 0.783]).

This benchmark does not turn cobidder status into legal proof. It answers the narrower validation question appropriate for a loser-side screen: whether the ranking concentrates firms adjacent to adjudicated cartel anchors at rates above what participation volume mechanically predicts. The temporal-holdout exercise in the next subsection extends the test to cases adjudicated after the BEC sample closes, where the labels are observed strictly after the score is constructed.

4.2. Temporal Holdout Discrimination

The broader CADE portfolio is more informative but less clean: several linked cases are adjudicated after the BEC sample closes. We treat it as a temporal holdout exercise

rather than as a pure contemporaneous benchmark. The labels are later-adjudicated, so the exercise is not a real-time field deployment. Its value is narrower: it asks whether a score built from earlier award records ranks firms whose loser-side cartel adjacency is revealed later by the adjudication record.

The holdout performance is strong. Computing the score from 2009–2016 participation and evaluating it on 2017–2019 labels gives firm-level AUC 0.864 (95% CI [0.858, 0.870]). The in-sample AUC is higher, 0.924, as expected. For an enforcement screen that would be run before the case file is complete, the holdout estimate is the more relevant benchmark.

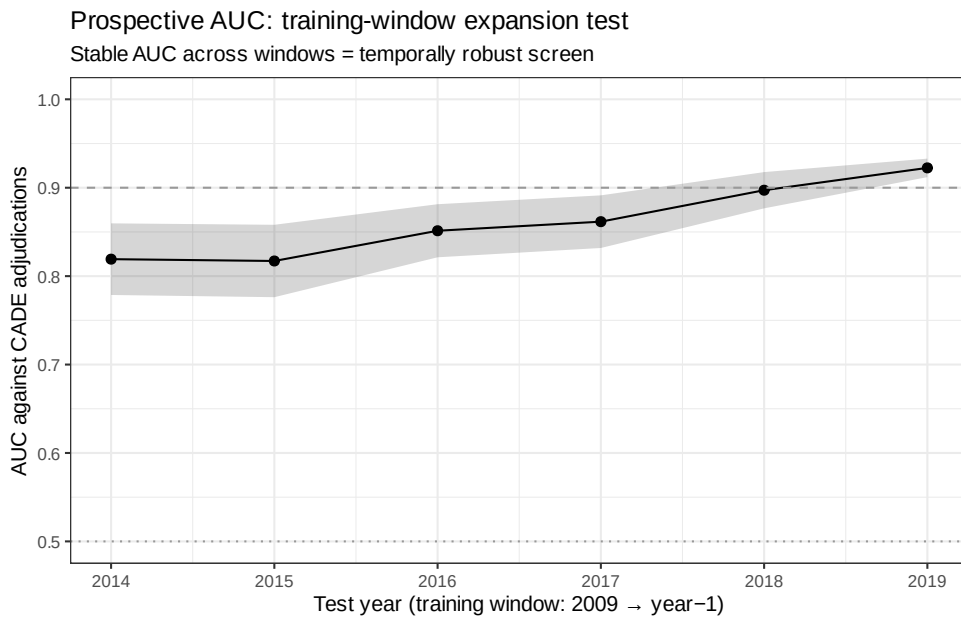


Figure 2: Temporal-holdout ROC curves. Scores are computed using participation observed before the test year; the test year supplies the out-of-time cobidder label.

Figure 2 reports the rolling-origin version of the same exercise: for each test year, scores are constructed using only prior participation. The relevant point is not that every year behaves identically, but that the ranking remains separated from random classification out of time.

The exercise supports temporal discrimination, not field deployment, causal prediction, or legal proof. It addresses retrospective fitting, since the score is fixed before the test-period labels exist, but it does not eliminate every timing concern: the labels are themselves observed later through adjudication, and that informational content is meaningful without being equivalent to real-time enforcement evidence. The next

subsection turns to a different threat: whether the discrimination is driven by participation-volume mechanics or by reuse of the same participation history.

4.3. *Participation Volume and Leakage*

The validation would be uninformative if it only reproduced its own construction. Two mechanical concerns matter. The first is *exposure*: firms that participate often have more opportunities to appear near any defendant, and the participation-stratified permutation in subsection 4.1 addresses that margin. The second is *leakage*: in item-level validation, the same firms can help construct both the participation score and the cobidder label. The two concerns are distinct and call for different audits.

We therefore do not use the in-sample item-level AUC of 0.995 as evidence of screening performance. That number is a reference value for the mechanical upper bound: it shows what happens before reuse of the same participation history is disciplined. The credibility of the validation comes from the audits that weaken that mechanical channel.

Two audits do the work. Holding cobidder firms out of score construction within cross-validation folds leaves AUC 0.891 (95% CI [0.887, 0.894]); this addresses reuse of the same firms in score and label construction. Computing the score only on 2009–2016 participation and evaluating it against 2017–2019 items leaves AUC 0.864 at the firm level; this addresses retrospective fitting by placing the labels out of time.

The drop from the in-sample reference is real and disclosed; it is the mechanical component. What remains is the relevant signal for the screening interpretation: the participation primitive continues to rank cartel-adjacent loser-side firms when labels are held out of score construction or placed out of time. The audits do not eliminate every source of mechanical overrepresentation, but they isolate the surviving signal from the channels they target. The next subsection turns to the opposite check: whether the same statistic also classifies direct CADE defendants, where a loser-side screen should perform worse.

4.4. *Direct-Defendant Asymmetry as a Scope Check*

The final validation check runs in the opposite direction. If the award-layer statistic were a generic cartel-membership detector, it should classify direct CADE defendants. A

loser-side ranking need not. The scope of the statistic predicts asymmetric performance: it is built from zero wins and repeated loser-side participation, while direct defendants are legal membership anchors that often occupy winner-side or allocation roles in the cartel environment.

That is what the data show. Against direct defendants in the broader BEC firm universe, AUC is 0.49, essentially random. We do not treat this as a hidden failure. It is a scope check: the statistic is not behaving like a general detector of cartel membership.

The interpretation is narrow. Direct defendants anchor adjudicated cartel environments; cobidders are the loser-side footprint the award layer can plausibly rank. Stronger performance against cobidders than against direct defendants is therefore not target shopping. It is the empirical signature of the screen’s stated scope.

The result rules out the wrong reading—that the screen detects cartel membership generically—and supports the right one: the screen ranks cartel-adjacent loser-side exposure around adjudicated cartel environments. It should be read together with the cobidder validation in the preceding subsections, where the same statistic concentrates on the population it was designed to rank.

Table 2: Validation benchmarks for the loser-side ranking. Each row addresses a distinct threat to interpretation.

Benchmark	Question	Result
Conservative CADE cases	Are frequent losers overrepresented among pre-2020 adjudicated cobidders, net of participation volume?	3.2× excess; AUC 0.748
Temporal holdout	Does pre-window participation rank later cartel-adjacent cobidders?	AUC 0.864
Leakage audit	Does signal survive when cobidders are held out of score construction?	Out-of-fold AUC 0.891
Direct-defendant scope check	Does the loser-side screen classify legal defendants symmetrically?	Direct-defendant AUC 0.49

Table 2 gives the compact view. The four rows are meant to be read together. The conservative benchmark anchors the label in adjudicated cases; the temporal holdout addresses retrospective fitting; the leakage audit owns the mechanical component; and the direct-defendant scope check keeps the legal object straight. Taken together, they support a limited but useful claim: award records can rank cartel-adjacent loser-side

firms before bid-level proof is recovered. They do not adjudicate membership, and they are not asked to.

5. Economic Profile of Cartel-Adjacent Losers

Section 4 validates the award-layer ranking against cartel-adjacent loser-side labels. It does not explain why those firms are adjacent to direct CADE defendants, and it does not turn adjacency into membership. This section asks a narrower profiling question: among firms already inside the frequent-loser stratum, do cobidders look different from other frequent losers in directions predicted by the screening interpretation?

The main alternative is mechanical adjacency. High-volume losing firms may appear near CADE defendants simply because they appear in many tenders. Section 4 addresses that concern statistically through participation-stratified permutation, holdout, and timing audits. Here we address it economically. We compare cobidders with frequent losers that do not co-bid with direct defendants. This is the hard comparison: both groups lose persistently and both cross the same participation threshold.

The evidence in this section is descriptive profile evidence, not adjudication. If cobidders differ from other frequent losers in deployment intensity, proximity to defendants, market concentration, and bid-level behavior, the cobidder label becomes harder to reduce to mechanical high-volume losing alone. The profile does not prove cover bidding in any particular tender, assign cartel membership, or establish liability. Its role is to discipline interpretation before the paper turns to the forensic sequencing exercises in Section 6.

5.1. Firm-Level Signature

The firm-level comparison is deliberately narrow. We compare cobidders with frequent-loser non-cobidders, not with the full BEC population. Both groups are always-losers, and both cross the same participation threshold. The exercise therefore does not ask whether cobidders look different from ordinary firms; it asks whether cartel-adjacent losers look different from other firms already selected by the screen.

We organize the comparison along four dimensions. *Deployment intensity* captures how broadly the firm is used in losing participation. *Cartel proximity* captures how often

its losing bids occur in tenders with direct CADE defendants. *Market specialization* captures whether activity is concentrated in narrower product spaces. *Repeat-pair density* captures whether losses cluster around the same winner relationships.

Cobidders move in the predicted direction on the first three dimensions. They participate more often than other frequent losers, 136.5 versus 76.7 participations (Cohen’s $d = +0.67$), and face more unique winners, 24.8 versus 13.5 ($d = +1.00$). They are also more proximate to legal cartel anchors: 1.5% of their loss-bids face a direct CADE defendant, compared with 0.2% for other frequent losers ($d = +0.46$). Their portfolios are more concentrated as well, with item-group HHI 0.380 against 0.288 ($d = +0.39$).

These patterns are inconsistent with a purely generic “all frequent losers are the same” reading. They support the idea that cobidders occupy a more targeted loser-side exposure profile than other firms already inside the frequent-loser stratum. The patterns do not prove cover bidding in any particular tender, assign cartel membership, or establish liability; they narrow the mechanical-adjacency alternative without closing every interpretation.

One measure moves in the wrong direction. Cobidders have a lower share of own loss-bids in repeat winner-pair clusters than frequent-loser non-cobidders, 21.6% versus 33.4% ($d = -0.38$). We do not use that measure as support for the cover-bidding interpretation. Its wrong sign is a useful discipline: the cobidder profile is not a sufficient statistic for cover bidding. The next subsection turns to bid-level signatures, where the LANCES bid records allow a complementary check on the same interpretation.

5.2. Bid-Level Signature

The LANCES export lets us ask whether the distinction between cobidders and other frequent losers also appears inside bids. This evidence is not part of the award-layer screen; it is a profile check using the richer forensic layer. For firms with at least five usable losing bids in convite or pregão, we compute the gap between each losing bid and the winning bid in the same tender-item, $(\text{bid} - \text{winning bid})/\text{winning bid}$, winsorized at $[-0.99, 10]$ before firm-level aggregation. The usable sample contains 182 cobidders and 2,369 frequent-loser non-cobidders.

The relevant comparison is not whether cobidders submit visibly uncompetitive bids. In modern cover-bidding environments, losing bids may need to remain plausible enough to preserve the appearance of rivalry. A cobidder can therefore bid closer to the winner than other frequent losers while still losing systematically. Higher within-firm dispersion in bid gaps is also consistent with a firm appearing in different losing roles across tenders. These patterns are informative but not uniquely diagnostic of cover bidding.

The data move in that direction. Cobidders bid closer to winners: their median bid gap is 0.582, compared with 0.809 for other frequent losers ($d = -0.281$). At the same time, their within-firm dispersion in bid gaps is higher, 1.207 versus 1.099 ($d = +0.147$). A logit of cobidder status within the frequent-loser stratum, controlling for $\log(1 + \text{tenders_count})$, preserves both signs: the median-gap coefficient is -0.708 ($z = -4.14$), and the dispersion coefficient is $+0.442$ ($z = +3.62$). The bid-level distinction therefore survives controlling for participation intensity.

The bid-level evidence has a limited but useful role. It shows that the cobidder distinction is not only another way of measuring high participation; it remains visible inside bids after the participation channel is held constant. The pattern is consistent with credible cover bidding and harder to reduce to mechanical high-volume losing alone. It is also compatible with some alternative economic mechanisms, including capacity-constrained bidding strategies and item-level price uncertainty. The evidence therefore narrows the mechanical-adjacency story; it does not prove that any specific bid was cover, establish agreement, or assign cartel membership.

Table 3: Cobidder signature within the frequent-loser stratum

Dimension	Cobidders	FL non-cobidders	Difference
Total participations	136.5	76.7	$d = +0.67$
Unique winners faced	24.8	13.5	$d = +1.00$
Loss-bids facing direct CADE defendant	1.5%	0.2%	$d = +0.46$
Portfolio item-group HHI	0.380	0.288	$d = +0.39$
Repeat winner-pair share	21.6%	33.4%	$d = -0.38$
Median bid gap to winner	0.582	0.809	$d = -0.281$
Within-firm SD of bid gap	1.207	1.099	$d = +0.147$

5.3. What the Profile Does and Does Not Establish

The profile evidence establishes a limited but important point. Cobidders are not simply indistinguishable high-volume losers. Within the frequent-loser stratum, they display greater deployment breadth, closer proximity to direct CADE defendants, more concentrated product portfolios, and bid-level patterns that survive controlling for participation intensity. These differences make the cobidder label harder to reduce to mechanical adjacency alone.

The profile does not establish legal or economic type in the strong sense. It does not prove that any specific bid was cover, that any cobidder was party to an agreement, or that the firm belongs to a cartel. It does not supply liability evidence or a damages base. The evidence is descriptive profile evidence: it supports the loser-side screening interpretation, but it does not adjudicate conduct.

The safest reading is therefore narrower than type identification. The profile is consistent with credible cover bidding—losing bids that remain plausible enough to preserve the appearance of rivalry while a firm cycles through cover roles across tenders ([Asker, 2010](#); [Marshall and Marx, 2012](#))—and harder to reconcile with a purely mechanical high-volume-loser interpretation. It is not uniquely diagnostic of cover bidding; alternative explanations remain available.

That is enough for the section’s role. The profile makes the cobidder target economically meaningful before the paper turns to the institutional question: whether an award-layer ranking can organize the more expensive bid-layer tools used in cartel forensics. Section 6 takes that step, asking whether placing the award-layer screen ahead of bid-layer forensics produces measurable savings in evidentiary effort without unacceptable losses in recall.

6. Sequencing Award-Layer Screening and Bid-Layer Forensics

The previous sections establish two points. Section 4 validates the award-layer ranking against cartel-adjacent loser-side labels; Section 5 shows that cobidders are not merely generic high-volume losers. The remaining question is institutional. What should an

agency do with a signal that is cheap enough to compute before bid-level evidence is recovered? The agency’s problem is not only prediction; it is evidence allocation under incomplete observability.

The answer is not to treat the award-layer screen as a substitute for bid analysis. The answer is to put it before bid analysis. Award-layer screening creates suspicion on the record layer the agency already observes; bid-layer forensics then investigates a smaller survivor pool using richer evidence from submitted bids. The object is therefore not a detector horse race under full observability. It is an evidence-allocation problem under incomplete observability.

This section evaluates the sequence in three steps. Subsection 6.1 asks how much ranking information is available at the award layer relative to the bid layer on the same cartel-adjacency target—an information-cost comparison, not a methodological dominance test. Subsection 6.2 asks whether award-layer and bid-layer signals are complements or substitutes. Subsections 6.3 and 6.4 ask whether the award-layer ranking can gate costly bid recovery while preserving much of the adjudication signal, both contemporaneously and under temporal holdout.

The third question is the operational object. The sequential gatekeeper exercise should be read as a cost-of-evidence result, not only as a classification table: Table 5 reports the reduction in bid-microdata pool size that the award-layer screen produces, alongside the recall it preserves against the adjudication record. That trade-off, not the comparison of stand-alone AUCs, is the institutional question this paper addresses.

6.1. Same Target, Different Information Costs

The forensic comparator is the bid-distribution tradition. Imhof–Wallimann-style detectors use within-tender bid moments—dispersion, skewness, spread, distance from the winning bid, and related features—to flag patterns consistent with coordinated bidding (Imhof, 2019; Wallimann et al., 2023). Those features live at the bid layer. They require per-bid offer values and enough usable bid histories to construct a firm-level forensic profile. The award-layer screen uses a thinner envelope: zero wins and participation counts. The question is not which detector is more sophisticated. It is how much ranking

information is available before the richer bid layer is opened.

Table 4 puts the two layers on the same cartel-adjacency target. That common target matters: it prevents the comparison from changing the object being ranked. What differs is the data envelope. The bid-layer benchmark requires per-bid offer values and within-tender moments; the award-layer measures require only routine award records. Table 4 is therefore an information-cost table before it is a classification table.

Within the always-loser pool for which bid-distribution features can be computed ($N = 16,779$), the binary frequent-loser flag reaches AUC 0.903 (95% CI [0.884, 0.923]). The continuous participation score reaches AUC 0.884 ([0.860, 0.908]). The full bid-layer benchmark reaches AUC 0.888 ([0.865, 0.911]). The combined award-layer and bid-layer specification reaches AUC 0.962 ([0.954, 0.969]).

This is not an outperformance claim. The award-layer statistic does not dominate bid-distribution forensics; the bid-layer benchmark remains the natural forensic tool when bid microdata are available. The point is sharper: a record-layer statistic, computed before forensic recovery of bid values, discriminates the same adjudication-anchored target in the same range as a much richer bid-layer benchmark. That information-cost result is the premise for sequencing. If the award layer is informative and not redundant with the bid layer—a question the next subsection addresses—it can decide where bid-layer forensics should begin.

Table 4: Discrimination at Different Information Costs

Model	Information required	AUC	95% CI
Frequent-loser flag	Award records: participation count and zero wins	0.903	[0.884, 0.923]
Participation intensity	Award records: $\log(1 + \text{tenders_count})$ and zero wins	0.884	[0.860, 0.908]
Bid-layer benchmark	Per-bid offer values and within-tender bid moments	0.888	[0.865, 0.911]
Award layer + bid layer	Participation intensity plus bid-distribution moments	0.962	[0.954, 0.969]

Notes: Five-fold cross-validated AUC against CADE-cobidder labels within the always-loser pool for which the bid-distribution benchmark can be computed ($N = 16,779$). The combined specification uses the same target and adds the award-layer score to the bid-layer moments.

6.2. Complementarity, Not Substitution

Comparable AUCs do not by themselves justify a two-stage architecture. If the award-layer statistic were merely a noisy proxy for bid-distribution moments, it would be useful mainly where bid microdata are unavailable. The stronger test is whether the award layer adds information once the bid layer is observed.

It does. Adding the award-layer statistic to the bid-layer benchmark raises discrimination to 0.962 ([0.954, 0.969]). In the identical same-sample audit, where all models are evaluated only on firms with complete bid-layer features, the frequent-loser flag adds +0.035 AUC over the full Imhof–Wallimann benchmark alone (DeLong $p = 0.014$). The combined specifications gain +0.096 — + 0.098 AUC over the bid-layer benchmark ($p < 0.001$).

The gain is substantively important because the two layers measure different legal-economic objects. Bid moments describe conduct inside tenders: dispersion, spread, rank distance, and other within-tender patterns. Participation intensity describes repeated loser-side exposure across tenders, before the bid file is opened. The first is closer to forensic evaluation; the second is closer to investigative priority. A regulator should not collapse them. The first creates a reason to look; the second helps evaluate what is found after looking.

The combined model is therefore a complementarity diagnostic, not the operational baseline. Running it on every firm requires full bid-layer observability before triage—precisely the cost the architecture is designed to manage. The enforcement use of the same non-redundancy appears in the next subsection: award-layer screening first creates the survivor pool, and bid-layer forensics then evaluates that pool.

6.3. Gatekeeping the Forensic Stage

The operational test is not whether the cheap screen beats the forensic benchmark. The relevant question is whether the cheap screen can reduce the forensic workload without discarding too much adjudication-anchored signal. Table 5 should therefore be read as a cost-of-evidence table before it is read as a classification table: each rule

is evaluated not only by precision and recall, but also by how many firms require bid microdata.

The baselines differ in observability. The award-layer-only rule uses no bid microdata. The bid-layer-only rule and joint scoring require bid microdata for the full evaluation pool. Joint scoring is an upper bound under full observability: it gives the classifier both award-layer and bid-layer features for every firm before triage has occurred. The sequential rule imposes the agency’s institutional constraint: it allocates bid-layer evaluation to a survivor pool selected by the award layer.

The sequential rule operates in two stages. First, it ranks firms using the award layer alone and keeps the top $K_1 = 2,000$. Second, it recovers or processes bid microdata only for those survivor firms, applies the bid-layer benchmark within the survivor pool, and selects the final top- k flags. The architecture therefore reserves the costly forensic step for the smaller set of firms the award layer already prioritizes; it does not assume bid microdata are available before triage.

The evaluation pool contains 11,676 always-loser firms with computable bid-distribution features and 193 CADE-cobidder positives. At the top 1,000 flags, joint scoring finds 142 cobidders, or recall 0.736, but requires bid microdata for all 11,676 firms. The sequential rule finds 131 of 193 cobidders, or recall 0.679, while requiring bid microdata for only 2,000 firms. Relative to joint scoring, the recall loss is 8%, while the bid-microdata footprint falls by 83%. Table 5 reports the same trade-off at the top-500 cutoff as well.

The sequential rule is not the precision champion—joint scoring has the easier job because it uses every information layer for every firm—and it is not claimed to be universally optimal. It earns its place by operationalizing the paper’s evidence-allocation problem: when bid recovery is costly, a modest loss in recall can be worth a large reduction in forensic-data exposure. That is the trade-off faced by an enforcement agency that cannot inspect every bidder, every bid, and every tender with the same forensic intensity.

Table 5: Gatekeeping the Bid-Layer Forensic Stage

Rule	k	TP	Precision	Recall	Bid-microdata firms
Award-layer only	500	66	0.132	0.342	0
Bid-layer only	500	73	0.146	0.378	11,676
Joint scoring, award + bid	500	112	0.224	0.580	11,676
Sequential: award $\rightarrow$ bid, $K_1 = 2,000$	500	96	0.192	0.497	2,000
Joint scoring, award + bid	1,000	142	0.142	0.736	11,676
Sequential: award $\rightarrow$ bid, $K_1 = 2,000$	1,000	131	0.131	0.679	2,000

Notes: Evaluation pool: 11,676 always-loser firms with computable bid-distribution features; positives are 193 CADE-cobidders. The bid-microdata column reports how many firms require per-bid offer values before the rule can be applied.

6.4. Temporal Holdout Gatekeeping

The in-sample gatekeeper exercise still deserves a timing check. A classifier trained and evaluated inside the full sample may rank firms well because it observes long histories close to the adjudicated outcome. A timing audit therefore asks whether the architecture still works when features are formed before the target window. This is not a real-time field deployment, but it imposes a stricter timing discipline on the evidence-allocation exercise.

We restrict feature construction to 2009–2016 and evaluate against the same post-window adjudication-anchored cobidder target. The temporal-holdout pool contains 8,257 always-loser firms with complete train-window bid features and 142 positives. The target itself remains adjudication-based and observed later; the discipline is on the score, not on the labels.

The audit preserves the architecture. At top 500, the award-only rule loses 47% of its in-sample true positives, joint scoring loses 24%, and the sequential rule loses only 9%. In levels, the sequential rule recovers 87 positives at top 500, compared with 85 for joint scoring. At top 1,000, it recovers 114 positives, compared with 111 under joint scoring.

This should not be read as proof that sequential gatekeeping dominates joint scoring. The safer interpretation is narrower and more useful: conditioning bid-layer forensics on an earlier award-layer screen remains competitive out of time while preserving a

smaller forensic footprint. The result reinforces the paper’s sequencing claim under timing discipline; it does not establish universal optimality, real-time enforcement success, or causal prediction.

The legal interpretation follows the same boundary as the empirical design. A frequent-loser flag cannot establish an agreement, identify a damages base, or prove that a particular bid was cover. It can identify where the agency should spend the next unit of investigative effort. The bid-layer stage then examines whether the bidding record contains the richer patterns needed for proof. For a journal concerned with how law works under institutional constraint, that sequencing is the contribution: a cheap administrative record can improve the allocation of costly legal evidence gathering without pretending to be legal evidence itself.

7. Scope, Limits, and Price Corroboration

The paper’s enforcement claim does not require a causal price effect. A screen can be useful if it ranks where forensic attention should begin, even when the price record is too coarse to support damages, markup estimates, or welfare losses. This section therefore does not treat price as proof of harm. It uses price evidence more narrowly: as descriptive corroboration and scope information.

The question is whether flagged environments look economically non-neutral, and where that interpretation stops. Broad-sample price associations can show whether the screen points toward procurement environments with higher negotiated prices. Overlap and segment results then discipline that reading. In particular, the sign reversal under overlap restriction is not a robustness failure to hide; it is the boundary that keeps price evidence from becoming a damages claim.

The section also records the paper’s portability and strategic-response limits. Award-layer triage is portable only where participant identities, winner identities, comparable item definitions, repeated procurement, and forensic follow-up are institutionally available. A static binary threshold can be gamed, which is why the architecture should be paired with threshold refresh, bunching checks, and bid-layer forensics. The purpose of this

section is therefore defensive and clarifying: it shows where the screen has economic content, and where the paper’s claims stop.

7.1. A Descriptive Price Imprint

The price regressions are reported to assess economic non-neutrality, not to estimate harm. If the award-layer screen ranks cartel-adjacent environments, it is useful to ask whether those environments also differ in the economic outcome agencies care about. The answer can only play a corroborative role. The estimates below are descriptive associations, not cartel markups, damages estimates, or causal effects of frequent-loser presence.

In the broad item-level price regressions, frequent-loser presence is positively associated with log negotiated unit price. The coefficient is 0.0677*** with item and year fixed effects and 0.0636*** after adding procuring-unit fixed effects. The association is larger in *pregão*, 0.0933***, than in *convite*, 0.0382**. Across the main descriptive estimators, the implied range is +3.6% – +7.7% (Table 6).

The *pregão*–*convite* split is useful as scope information, not causal identification. If frequent losers were mainly quorum fillers created by the three-bidder rule in *convite*, the price imprint should be stronger there. Instead, the larger estimate appears in *pregão*, the real-time electronic environment where loser-side positioning is easier to deploy repeatedly. This contrast supports the screening interpretation, but it does not identify a procedure-specific mechanism.

The boundary is important. The positive broad-sample association is not a causal estimate of cover bidding’s effect on prices. The available quasi-experimental routes do not support that claim: discontinuity designs around procedure thresholds lack the required density behavior, and the policy-change design does not supply credible parallel trends. The role of the price evidence is therefore limited. It shows that flagged environments are economically non-neutral in the broad record; the next subsection shows where that interpretation stops.

Table 6: Price Corroboration and Scope

Exercise	Estimate	Standard error / p	Interpretation
Broad sample, item and year FE	0.0677***	0.0230	Positive price imprint
Broad sample, item, year, PBU FE	0.0636***	0.0215	Positive after buyer controls
Pregão subsample	0.0933***	0.0255	Larger in electronic auctions
Convite subsample	0.0382**	0.0186	Smaller under quorum-prone procedure
Overlap-cell ATT	-0.097	$p < 0.001$	Aggregate sign reversal
PS-trimmed ATT	-0.307	—	Stronger negative under trimming
Tender-value Q4, overlap ATT	+0.041	$p = 0.045$	Positive in high-value segment
Direct-CADE overlap items	-0.061	$p = 0.45$	Statistically null

Notes: Dependent variable is log negotiated unit price. The broad-sample rows use the main BEC item-level sample. The overlap rows restrict or reweight comparisons to cells with treated and untreated support. Estimates are descriptive and are not interpreted as causal price effects, markups, or damages calculations.

7.2. Where the Price Evidence Stops

The overlap results define where the price evidence stops. The sign reversal is not a robustness failure to hide; it is the boundary that keeps price evidence from becoming a damages claim. The broad-sample and overlap estimates answer different questions over different parts of the procurement distribution, and the difference is what disciplines the interpretation.

The broad-sample estimate averages over the environments into which frequent losers actually sort. The overlap estimate restricts or reweights toward cells where frequent-loser-present and frequent-loser-absent items have comparable support on observables. These are different descriptive objects. Neither is a causal effect: both are reweighted averages of the same conditional association under different sampling weights.

Under overlap-cell ATT reweighting, the aggregate coefficient reverses to -0.097 ; under propensity-score trimming it becomes -0.307 . Only 1.06% of treated items lack a within-cell counterfactual, so the reversal is not mainly a dropped-observation story. It is a reweighting story: the broad estimate and the overlap estimate put weight on different parts of the item-value distribution.

The segment decomposition makes that point explicit. In the lowest three tender-value quartiles, the overlap estimates are negative. In the highest tender-value quartile, the estimate remains positive, $+0.041$ ($p = 0.045$). That segment is where cartel rents are most plausible ex ante, because the same fixed cost of coordination is spread over larger

contracts. The interpretation remains limited: the high-value result is segment-level scope evidence, not a general price effect.

The direct-CADE overlap estimate is also null, -0.061 ($p = 0.45$), which reinforces the same boundary. The price regressions do not identify a treatment effect, a markup, or a damages base. They show that the award-layer screen points to economically non-neutral environments in parts of the distribution, while the overlap reversal prevents that evidence from carrying the paper’s main claim. The next subsection turns from descriptive price evidence to the institutional conditions under which the screening architecture is portable.

7.3. Portability Conditions

The architecture is more portable than the estimates. The BEC results come from a particular procurement platform, legal environment, and product mix; they should not be read as numerical external validity for other jurisdictions. What can travel is the information problem itself: an agency may routinely observe an award layer while recovering bid-level evidence only through costly forensic process.

A candidate setting needs several institutional primitives. Participant identities and winner identities must be observed. Item definitions must be stable enough for repeated participation to be meaningful. Procurement must be repeated enough to generate participation histories. Persistent losing must carry economic content rather than being purely mechanical. And the bid layer must be recoverable for forensic follow-up, because the award layer only creates suspicion; it does not supply proof.

Those conditions are more plausible in standardized, repeated, electronically recorded procurement than in bespoke services, sparse infrastructure projects, or systems with incomplete participation records. BEC fits the favorable profile particularly well: standardized goods are common, electronic records are routine, and pregão-style auctions preserve the loser-side interaction environment in which the signal is strongest.

The architecture is harder to deploy where one or more primitives fail. Bespoke services, large one-off infrastructure contracts, and sparse procurement environments may not generate the participation history the screen requires. Systems with incomplete loser records, weak item comparability, or no feasible bid-layer recovery would create

suspicion without a path to proof. Under those conditions, award-layer triage offers little institutional value.

Portability therefore requires recalibration. Another agency should not import the BEC threshold or item definitions mechanically. It should rebuild the participation history, define comparable procurement categories, choose a capacity-constrained threshold or top- k rule, and validate against locally meaningful enforcement labels. The portable lesson is the sequencing architecture, not the number 14. The next subsection turns to the second adaptation question: how the rule responds to firms that know it exists.

7.4. Strategic Response

Strategic response is a real limit on any administrative screen. Once firms understand that persistent zero-win participation can trigger attention, they may rotate cover roles, engineer occasional wins, split identities, or cap participation below a known threshold. The question is not whether the binary rule is manipulation-proof—it is not—but how fragile different parts of the architecture are under different adaptation strategies.

The adaptation simulation shows a clear hierarchy. Simple rotation leaves AUC essentially unchanged, 0.931 versus baseline 0.928. Occasional wins also leave AUC nearly unchanged, 0.929. CNPJ splitting is more damaging, 0.891. Threshold-aware capping is more damaging still, 0.801. Combining capping with rotation lowers AUC to 0.632.

The pattern is informative. Simple rotation and occasional wins are absorbed by the persistence signal; identity fragmentation begins to break the participation history; behavior targeted at the static cutoff is the most damaging. That is exactly why the number 14 should be treated as operational rather than structural. It is useful for an auditable first-stage implementation in this setting, not as a permanent enforcement boundary.

The practical implication is adaptive deployment. Agencies should refresh thresholds, monitor bunching just below the rule, use continuous ranks or capacity-constrained top- k queues, and pair award-layer screening with bid-layer forensics whenever bid files can be recovered. The screen creates a moving queue for forensic attention, not a fixed legal trigger.

The simulation is a diagnostic, not a full equilibrium model of evasion. It shows how the ranking can degrade under simple adaptation strategies; it does not solve the strategic problem at its root. The paper’s claim therefore remains limited: award-layer triage can help allocate forensic attention, but it requires adaptive implementation. Strategic response weakens naive cutoff use more than it invalidates the two-stage architecture.

8. Conclusion

Cartel enforcement often begins before the agency has the evidence it would need to win a case. Procurement records then pose an institutional question that is earlier than proof: can the cheap layer help decide where the expensive layer should be opened? This paper studies that question on São Paulo’s BEC platform. The answer is yes, but only within a disciplined legal and empirical scope.

The award-layer statistic ranks firms on persistent zero-win participation. That ranking is not a membership label. It is a loser-side exposure measure built from records an agency can query before recovering bid-level evidence. Validation therefore targets cartel-adjacent cobidders rather than direct-defendant status. The ranking survives a conservative adjudication benchmark, participation-volume discipline, temporal holdout, leakage audits, and the direct-defendant scope check that rules out the wrong interpretation of the screen as a generic cartel-membership detector.

The profile evidence gives the target economic content without turning it into proof. Cobidders inside the frequent-loser stratum differ from other frequent losers in deployment breadth, proximity to direct defendants, product concentration, and bid-level behavior. These patterns are consistent with the loser-side screening interpretation and harder to reduce to mechanical high-volume losing alone. They do not prove cover bidding, agreement, membership, or liability.

The main contribution is the sequencing result. Bid-distribution forensics remains necessary, but it need not be the first stage. Used as a gatekeeper, the award-layer ranking reduces the bid-microdata pool for the forensic stage by 83% while still recovering 131 of 193 adjudicated cobidders. Joint scoring is the full-observability upper bound; sequential gatekeeping is the operational architecture for agencies facing costly bid recovery.

The limits are part of the result. The price evidence is corroborative, not a damages calculation. The architecture is portable only where participant identities, winner identities, repeated comparable procurement, meaningful losing participation, and recoverable bid files coexist. A static cutoff can be gamed, so implementation requires refreshed thresholds, bunching checks, continuous or top- k queues, and bid-layer follow-up.

Award-layer triage is therefore not a shortcut to liability. It is a way to order suspicion under incomplete information. The statistic identifies where legal attention should start, not where legal responsibility ends.

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